MIL-C-82284C
1 February 1984
SUPERSEDING
MIL-C-82284B(NU)
15 May 1979

MILITARY SPECIFICATION

COVERALLS, FIREMEN'S, ALUMINIZED, PROXIMITY

This specification is approved for use by all Departments and Agencies of the Department of Defense.

1. SCOPE

1.1 Scope. This specification covers the requirements for a heat protective coverall constructed of highly reflective aluminized aramid fabric with a chloroprene coated liner.

1.2 Classification. The coveralls covered by this specification shall be of one type only and in the following sizes (see 6.2):

Small/Medium - (37-41)
Large/X-Large - (42-46)

2. APPLICABLE DOCUMENTS

2.1 Government documents.

2.1.1 Specifications, standards, and handbooks. Unless otherwise specified, the following specifications, standards, and handbooks of the issue listed in that issue of the Department of Defense Index of Specifications and Standards (DoDISS) specified in the solicitation form a part of this specification to the extent specified.

Beneficial comments (recommendations, additions, deletions) and any pertinent data which may be of use in improving this document should be addressed to: Officer in Charge, Navy Clothing and Textile Research Facility, 21 Strathmore Road, Natick, MA 01760 by using the self-addressed standardization Document Improvement Proposal (DD Form 1426) appearing at the end of this document or by letter.

FSC 8415

Source: https://assist.dla.mil -- Downloaded: 2026-10-26T04:40Z
Check the source to verify that this is the current version before use.

SPECIFICATION

FEDERAL

V-F-106	- Fasteners, Slide, Interlocking
FF-R-556	- Rivets, Solid, Small; Rivets, Split, Small; Rivets, Tubular, Small; and Caps, Rivets; General Purpose
FF-W-92	- Washers, Metal, Flat (Plain)
KK-L-2004	- Leather, Cattlehide, Deerskin and Horsehide, Chrome-Tanned
QQ-N-290	- Nickel Plating (Electrodeposited)
QQ-P-416	- Plating, Cadmium (Electrodeposited)
DDD-L-20	- Label; For Clothing, Equipage and Tentage, (General Use)
DDD-T-86	- Tape, Textile, Cotton, General Purpose (Unbleached, Bleached or Dyed)

MILITARY

MIL-B-371	- Braid, Textile, Tubular
MIL-C-3735	- Cuffs, Knit, Wrist and Ankle, and Cloth, Knitted
MIL-B-17757	- Boxes, Shipping, Fiberboard (Modular Sizes)
MIL-C-19699	- Cloth, Coated, (Nylon Taffeta)
MIL-T-43636	- Thread, Nylon, Non-Melting
MIL-T-44100	- Thread, Para-aramid, Spun, Intermediate Modulus
MIL-C-87076	- Cloth, Coated, Aramid, Aluminized

STANDARDS

FEDERAL

FED-STD-311	- Leather, Methods of Sampling and Testing
FED-STD-751	- Stitches, Seams, and Stitchings

MILITARY

MIL-STD-129	- Marking for Shipment and Storage
MIL-STD-1666	- Classification of Defects for Evaluating Quality of Aluminized Firemen's Clothing

(Copies of specifications, standards, drawings, and publications required by contractors in connection with specific procurement functions should be obtained from the procuring activity or as directed by the contracting officer.)

2.1.2 Other Government documents, drawings, and publications. The following other Government documents, drawings, and publications form a part of this specification to the extent specified.

LAWS AND REGULATIONS

US POSTAL SERVICE MANUAL

Source: https://assist.dla.mil -- Downloaded: 2026-10-26T04:40Z
Check the source to verify that this is the current version before use.

(Copies of manual may be obtained from the Superintendent of Documents, US Government Printing Office, Washington, DC 20402).

2.2 Other publications. The following documents form a part of this specification to the extent specified herein. The issues of documents indicated as DOD adopted shall be the issue listed in the current DoDISS or supplement thereto, if applicable.

NATIONAL MOTOR FREIGHT TRAFFIC ASSOCIATION, INC., AGENT

National Motor Freight Classification

(Application for copies should be addressed to American Trucking Association, Inc., Traffic Department, 1616 P Street, NW, Washington, DC 20036).

UNIFORM CLASSIFICATION COMMITTEE, AGENT

Uniform Freight Classification

(Application for copies should be addressed to Uniform Classification Committee, Room 1106, 222 South Riverside Plaza, Chicago, IL 60606.)

3. REQUIREMENTS

3.1 Guide sample. Guide samples, when furnished, are solely for guidance and information to the contractor (see 6.3). Variation from this specification may appear in the sample, in which case this specification shall govern.

3.2 First article approval. When specified (see 6.2), the contractor shall furnish sample unit(s) for first article inspection and approval (see 4.3).

3.3 Materials.

3.3.1 Basic material. The basic material for the coveralls shall be lightweight aluminized aramid fabric conforming to type II of MIL-C-87076.

3.3.2 Liner. The material for lining the outershell shall be coated nylon taffeta conforming to type II of MIL-C-19699.

3.3.3 Leather. The leather strips for reinforcing the bottom of the sleeves and legs shall be a flesh split cattlehide, natural color of tannage, of not less than 2 3/4 nor more than 3 1/2 ounce thickness conforming to type IV, class 2 of KK-L-2004 after silicone treatment specified in 3.3.3.1, except for the following:

a. Shrinkage temperature shall be not less than 105° C.
b. Tests for percent ash and chloroform-soluble material not required.

Source: https://assist.dla.mil -- Downloaded: 2026-10-26T04:40Z
Check the source to verify that this is the current version before use.

3.3.3.1 Silicone treatment. The leather shall be treated with an approved durable silicone compound to lower its water absorption (see 4.4.1, 6.2 and 6.4).

* 3.3.4 Thread. The thread for assembling the coveralls shall be an intermediate modulus, spun para-aramid thread conforming to MIL-T-44100 size T-50. Overedging operations may be performed using size T-35. As an alternate, a non-melting nylon (Aramid) thread, natural shade, conforming to size B of MIL-T-43636, may be used.

3.3.5 Cuffs, wrist. The material for the cuffs shall be knitted wool in black, olive drab or natural and shall conform to type I, class 2, size 1 of MIL-C-3735 except that only colorfastness to crocking and perspiration shall apply when cuffs are dyed.

3.3.6 Slide fastener. The slide fastener for the front closure shall conform to type I, style 3, size MH, with wire stirrup or long tab pull and thong of V-F-106, except that all metal components of the slide fastener shall be fabricated of brass and purchased from the same supplier. The thong shall be constucted of leather specified in 3.3.3, cut in a strip at least 1/4 inch wide and 4 inches long. The slider may be zinc alloy brass plated. The fastener tape shall be cotton, natural color or dyed olive drab. Dyed tape shall show colorfastness to laundering only. The ends of the tape shall extend approximately 1 1/2 inches beyond the ends of the chain. The chain length shall finish 25 1/2 (+ 1/4) inches for size Small/Medium (37-41) coverall and 26 1/2 (+ 1/4) inches for size Large/Extra Large (42-46) coverall.

3.3.7 Tape, cotton. The cotton tape for reinforcing the crotch seam, inseams, side seams, and sleeve seams shall conform to type IV, class 1 or 2, 1 1/4 inches wide of DDD-T-86, except that the requirement for shrinkage shall not apply.

3.3.8 Hardware.

3.3.8.1 Snaps and dee rings. The snaps and dee rings for fastening the outside protective flap shall be cast malleable iron, brass or cast aluminum conforming to the requirements specified in figure 2 (see 4.4.1). The malleable iron shall be cadmium plated and shall conform to type I, class 3 of QQ-P-416. The brass shall be nickel plated and shall conform to class 2 of QQ-N-290, except that the minimum thickness of the plating shall be 0.0005 inch.

3.3.8.2 Rivets, tubular with caps or washers. The rivets for attaching the snaps and dee rings shall conform to type XII, class 3, grade B or F of FF-R-556, except that they shall be drilled to the head or partially drilled to the head and the inside of the rivet shank shall be chamfered. The rivets shall be 9/64 inch nominal diameter and their length shall be sufficient to assure proper clinching. The caps shall conform to type II of FF-R-556. The brass rivets and caps shall be nickel plated to conform with class 2 of QQ-N-290, except that the minimum thickness of the plating shall be 0.0005 inch. An aluminum washer that conforms to type A, grade II, class C, no. 8 of FF-W-92 may be used in lieu of caps for securing the tubular rivets.

3.3.9 Labels.

Source: https://assist.dla.mil -- Downloaded: 2026-10-26T04:40Z
Check the source to verify that this is the current version before use.

3.3.9.1 Identification label. Each coverall shall have a combination identification and size label conforming to type VI, classes 1 and 2 of DDD-L-20, except that colorfastness requirements shall not apply. The label shall bear the following inscription:

COVERALLS, FIREMEN'S, ALUMINIZED, PROXIMITY
BASE FABRIC: ARAMID (100%)
CONTRACT NO: DLA-100-00-0-0000 (EXAMPLE)
STOCK NO: 8415-00-000-0000 (EXAMPLE)
SIZE: SMALL/MEDIUM (37-41) (EXAMPLE)
NAME OF CONTRACTOR:

3.3.9.2 Instruction label. Each coverall shall have an instruction label conforming to type VI, class 3 of DDD-L-20, except that colorfastness requirements shall not apply. The label shall bear the following inscription:

INSTRUCTIONS

1. Coveralls are sized to wear heavy clothing underneath for added protection against heat penetration.
2. Hang to dry and store properly to prevent damage to aluminized surface.
3. Clean by gently rubbing surface with cloth using mild soap, PERCHLORETHYLENE or ISOPROPANOL.
4. Do not use aluminized paint to repair worn out surface as it will not provide protection.
5. To maintain reflective properties and to extend service life, use coveralls for intended purposes only.

NOTE: Aluminized coating is damaged by contact with fire extinguishing agent Potassium Bicarbonate Powder (PKP). Before storing garment with PKP residue present, the following is recommended:

1. Dry powder should be removed with clean soft brush or blower.
2. Wet powder should be removed by thoroughly flushing with water and wiping the surfaces down with a clean soft cloth. Damp or wet garments should be hung to dry.

3.3.10 Repair of aluminum film. The aluminum film used for patches for repairing uncoated or cracked areas on the finished coverall shall be the same as specified for coating the aramid base cloth. The area to be repaired shall not exceed 1 inch in the longest direction. The patch shall be completely bonded onto the aluminum film of the coverall and shall be so placed that it extends not less than 1/2 inch beyond all edges of the uncoated or cracked area. No coverall shall contain more than 3 patches on single ply areas or 3 patches on multiple ply areas (total of 6 patches). The patch shall not exceed 2 inches in any direction. The unsupported aluminum film and self-curing adhesive used for repairing shall be obtained from the basic material supplier or their source (see 3.3.1).

* 3.4 Design. The coveralls shall be fabricated from aramid material having a highly reflective aluminized outer surface and a waterproof, flame retardant, chloroprene coated liner. The coveralls shall have a bellows back, knitted wool wristlets, and leather bound sleeve and leg openings. The front entry shall consist of a slide fastener closure and be covered by a protective flap secured with four snaps and dee rings (see figures 1 and 2).

Source: https://assist.dla.mil -- Downloaded: 2026-10-26T04:40Z
Check the source to verify that this is the current version before use.

3.4.1 Figures. If inconsistencies exist between the written specification and the figures, the written specification shall govern.

* 3.5 Patterns. Standard patterns will be furnished by the Government to the contractor for use in cutting the working patterns (see 6.3). The patterns, except as otherwise indicated in Table I, have the following seam allowances:

Centerback seam	- 1/2 inch
Bellows to back	- 3/8 inch
Shoulders	- 1/2 inch
Neck facings to neckline	- 3/8 inch
Sleeve inseams, leg inseams, and outseams	- 1/2 inch
Sleeves to armholes	- 1/2 inch

The working patterns shall be identical to the Government patterns. Neither the Government patterns nor the working patterns shall be altered in any way.

* 3.5.1 List of pattern parts. The component parts of the coveralls shall be cut from the material as specified below:

Material	Nomenclature of pattern parts	Cut parts
Basic Material	Front	2
	Back	2
	Sleeve	2
	Outside protective flap (right side)	2
	Protective inside flap (left side)	1
	Back inverted pleat facing	1
	Back neck facing	1
	Front neck facing	2
	Hanger loop	1
Liner material	Front	2
	Back	2
	Sleeve	2
	Inside protective flap liner	1
	Inside armhole hanging tabs	8

3.6 Construction.

3.6.1 Stitches, seams, and stitchings. Stitch, seam, and stitching types specified in Table I shall conform to FED-STD-751. Whenever two or more methods of seams or stitches are given for the same part of an operation, any one may be used. Where stitch type 401 is used, the looper (underthread) shall be on underside of the assembly. In order to minimize the size of the needle hole in the fabric, the needle used for the seaming operations shall be of the smallest size which will accommodate the specified size of thread. Wherever LSq-2 seam type is specified, the top ply shall not be turned back sharply over the first row of stitching, but turned in such a manner that a loose welt is formed.

Source: https://assist.dla.mil -- Downloaded: 2026-10-26T04:40Z
Check the source to verify that this is the current version before use.

3.6.2 Thread breaks and ends of seams. Ends of all seams and stitchings when not caught in other seams or stitching shall be backstitched or overstitched not less than 3/8 inch. The ends of a continuous line of stitching shall be overlapped not less than 1/2 inch. Thread breaks or skipped stitches (301 or 401 stitch types) may be repaired by 301 stitch type with the stitching anchoring both ends of the break or skipped stitches not less than 1/2 inch.

3.6.3 Stitches per inch. The minimum and maximum number of stitches per inch shall be as specified in Table I.

3.7 Manufacturing operations requirements. The coveralls shall be manufactured in accordance with the operation requirements specified in Table I. The contractor is not required to follow the exact sequence of operations listed provided that the finished coveralls are identical to those produced by following the sequence of operations as listed in Table I.

3.7.1 Size marking. The component parts of the coverall shall be bundled, marked or ticketed to assure proper size assembly. The use of metal fastening devices or sewn-on tickets for marking is prohibited. Adhesive tickets, when used, shall not discolor the material and the adhesive mass shall not adhere to the material upon removal of the tickets.

NOTE: The use of ink pad numbering machine, rubber stamp or pencil is allowed, provided the numbering does not show on the outside of the coverall and wherever possible, is covered by the seam allowance.

Source: https://assist.dla.mil -- Downloaded: 2026-10-26T04:40Z
Check the source to verify that this is the current version before use.

TABLE I - CONSTRUCTION OF COVERALLS

No.	Operation	Stch Type	Seam/ Stch type	Stch/ in	Thread (3.3.4)
1.	Cutting.				
	a. Cut the coveralls in strict accordance with patterns which show size, placement of component parts, directional lines for cutting, and notches and marks for proper assembly. Drill holes are prohibited, except for placement of the snaps. Facings and protective flaps may be cut in the warp or filling direction. Measurements taken from the top and the bottom of body and sleeve directional lines to the selvage edge of the fabric may have a maximum difference of 2 1/2 inches.				
2.	Replacement of damaged parts.				
	During the spreading, cutting and manufacturing operation, components having material defects or damages that are classified as defects in MIL-STD-1666 shall be removed from production and replaced with nondefective and properly matched components.				
3.	Size marking (see 3.7.1).				
	Mark, bundle or ticket the component parts to assure proper size assembly.				
4.	Component assembly.				
	a. Armhole/underarm and crotch seams shall not be staggered more than 5/8 inch.				
	b. The aluminized side of the basic material shall be the face and shall finish as the exposed surface of the coverall.				
	c. The uncoated side of the liner material shall be the face, and shall be the exposed surface on the inside of the coverall. Silicone spray may be used on the coated surface to facilitate stitching operations.				

Source: https://assist.dla.mil -- Downloaded: 2026-10-26T04:40Z
Check the source to verify that this is the current version before use.

TABLE I - CONSTRUCTION OF COVERALLS

No.	Operation	Stch Type	Seam/ Stch type	Stch/ in	Thread (3.3.4)
4.	Component assembly (cont'd)				
	d. Joining seams on the outer shell shall be lapped as follows: 1. Backs over fronts at shoulders 2. Body over sleeves at armholes 3. Left back over right back at centerback 4. Fronts over backs at side seams/ underarm seams and inseams				
	e. Leather strips - In lieu of a one-piece binding, strips may be spliced with not more than one splice per binding. When so cut, the ends of the strips shall be joined with 1/4 to 3/8 inch seam allowance (see operation 20).	301	SSa-1	6-8	
5.	Join center back seam and attach bellows piece (outer shell).				
	a. The back shall be two sections with a center back bellows piece that extends from above the waistline to below the neck opening, in accordance with marks on patterns.				
	b. Join the edges forming the center back seam from the neck edge to the upper mark and from the lower mark to the crotch. The distance between the marks is the bellows opening. Slash the right seam allowance to the stitching at the base of the bellows extension.	301 or 401	SSa-1	6-8	
	c. Spread open seam allowance above opening, fold back the edges of the bellows opening, and stitch 1/4 inch gauge on each side of the seam. Continue stitching along the folded edge of the bellows opening. At the base of the opening, turn seam allowance to the left, and continue stitching on the left side only to the crotch.	301	SSz-3(b), EFa-1 and LSq-2(b)	6-8	
	d. Position the bellows piece on the left and right folded sections of bellows opening face to face. Stitch 3/8 inch from raw edges on each side.	301 or 401	SSa-1	6-8	

Source: https://assist.dla.mil -- Downloaded: 2026-10-26T04:40Z
Check the source to verify that this is the current version before use.

TABLE I - CONSTRUCTION OF COVERALLS

No.	Operation	Stch Type	Seam/ Stch type	Stch/ in	Thread (3.3.4)
5.	Join center back seam and attach bellows piece (outer shell). (cont'd)				
	e. Stitch top and botton of bellows piece through all plies, 3/8 inch from the top and bottom edges across the width of the bellows piece.	301	SSbd-1	6-8	
6.	Prepare fronts (outer shell).				
	a. Stitch the front darts according to pattern marks.	301 or 401	OSf-1	6-8	
	b. Stitch 1/2 to 5/8 inch from raw edge, face to face, the fronts from crotch to step edge. Slash seam allowance to facilitate tape stitching.	301 or 401	SSad-3(a)	6-8	
	c. Spread crotch seam open and flat, and straddle with a 1 1/4 inch wide cotton herringbone tape. Stitch 3/8 inch from the seam on each side through all plies.	301 or 401	SSad-3(b)	6-8	
7.	Join shoulders (outer shell).				
	a. Join, face to face, the fronts and back at each shoulder. Turn and stitch 1/4 to 3/8 inch from folded edge. Or, a single lapped, 1/4 to 3/8 inch gauge double stitched seam may be used.	301 or 401	LSq-2 or LSb-2	6-8	
8.	Assemble and attach outside protective flap.				
	a. Stitch, 3/8 inch from raw edge, face to face, the front, top and bottom edges of the protective flap (right side) to the bottom ply. Trim corners, turn to finished position forcing out corners and edges, and stitch 3/4 inch from the front edge; 3/8 inch from the top and bottom edges.	301 or 401	SSe-2	6-8	
	b. Position and stitch 3/8 inch from raw edge of the flap to the right front as indicated by pattern marks, outsides together. Turn flap to finished position and stitch 1/2 inch from the folded edge.	301	LSbl-2	6-8	
	c. The top edge of the protective flap shall finish 7/8 (± 1/8) inch below the raw edges of the neckline.				

Source: https://assist.dla.mil -- Downloaded: 2026-10-26T04:40Z
Check the source to verify that this is the current version before use.

TABLE I - CONSTRUCTION OF COVERALLS

No.	Operation	Stch Type	Seam/ Stch type	Stch/ in	Thread (3.3.4)
9.	Set sleeves (outer shell).				
	a. Join, face to face, the sleeves to armholes, matching notches. Turn to finished position and stitch 1/4 to 3/8 inch from folded edge. Or, a single lapped, 1/4 to 3/8 inch gauge double stitched seam may be used.	301 or 401	LSq-2 or LSb-2	6-8	
10.	Join sleeve inseams, outseams and leg inseams (outer shell).				
	Join seams face to face. Turn to finished position and stitch 1/4 to 3/8 inch from folded edge. Or, a single lapped, 1/4 to 3/8 inch gauge double stitched seam may be used.	301 or 401	LSq-2 or LSb-2	6-8	
11.	Attach snaps to left front (outer shell).				
	a. The four snaps shall be positioned as follows:				
	1. With the center fronts aligned and the flap laying smoothly over the fronts, snaps shall finish horizontally on the left front with the snap heads adjacent to the edge of the flap.				
	2. The top and bottom snaps shall finish 1 3/8 (± 1/8) inches from the top and bottom edges of the flap.				
	3. Space the remaining two snaps evenly in-between.				
	b. Set the snaps with three rivets reinforced on the underside with a basic fabric patch that extends not less than 1/4 inch beyond the caps or washers. The heads of the rivets shall finish on the outside.				
	NOTE: The snaps shall be attached through the outer shell only.				

Source: https://assist.dla.mil -- Downloaded: 2026-10-26T04:40Z
Check the source to verify that this is the current version before use.

TABLE I - CONSTRUCTION OF COVERALLS

No.	Operation	Stch Type	Seam/ Stch type	Stch/ in	Thread (3.3.4)
18.	Assemble liner and outershell.				
	a. Stitch top edge of the facing to the outer shell neckline, face to face, 3/8 inch from edges, matching center backs. The zipper tape/facing seam shall be turned away from center front and stitching shall continue across the top of the zipper tape. The extension of the left front shell beyond the zipper shall match the extension of the right front within 3/8 inch.	301	SSe-2(a)	6-8	
	b. Notch seam allowance and turn the neck facing/liner assembly to finished position. Join the folded ends of the anchor strips in the liner to the folded ends of the corresponding anchor strips in the outer shell by bartacking or stitching and backstitching about 1/4 inch from the edges.	Bartack or 301	SSa-1	22-28 per bartack or 12-16	
	c. Fold under the extended front edges of the outershell so that the folds are even with the outer edges of the zipper teeth, and edgestitch 1/4 inch from folded edge of the neckline all around.	301	SSe-2(b)	6-8	
	d. Stitch down the fronts, 1/2 inch from the folded edges, catching the left and right halves of the zipper tape. Connect the two rows of stitching at the base of the zipper.	301	SSs-1	6-8	
	e. Stitch a second row of stitching around the neckline 1 inch from the neck edge starting at the left front edge and continuing around to the flap on the right front. Stitching shall secure the facing to the outer coverall shell.	301	--	6-8	
	f. Stitch liner to outershell at leg and sleeve bottoms about 1/4 inch from the edges.	301 or 401	SSa-1	6-8	
19.	Make and attach inside protective flap and dee-rings.				
	a. Center the instruction label on the uncoated side of the inside protective flap liner approximately 1 1/2 inches below the top raw edge, and stitch all four sides 1/16 inch from the edges.				

Source: https://assist.dla.mil -- Downloaded: 2026-10-26T04:40Z
Check the source to verify that this is the current version before use.

TABLE I - CONSTRUCTION OF COVERALLS

No.	Operation	Stch Type	Seam/ Stch type	Stch/ in	Thread (3.3.4)
19.	Make and attach inside protective flap and dee-rings. (cont'd)				
	b. Stitch 3/8 inch from the raw edges the top, bottom and the inside edges of the protective flap face to face with the liner. Trim corners, turn and topstitch 3/8 inch gauge. Overedge raw edges together.	301 or 401	SSe-2;	6-8	
		503, 504 or 602	SSa-1		
	c. Position the aluminized side of the flap against the inside of the coverall at left front. Overlap the front about 1 inch with the top finishing 1/2 (± 1/8) inch below the neck edge, even with the top of the outside flap. Stitch flap to coverall about 1/4 inch from the overedged edge, parallel to the previously formed row of stitching on the coverall.	301	LSa-1		
	d. Rivet four dee-rings through the front edge of the outside protective flap to correspond with the snaps on the left front. The back inside edges of the dee-rings shall be at the edge of the flap, and when fastened, there shall be no twisting or bulging of the fronts.				
20.	Attach leather strips.				
	Insert a two-inch wide leather strip on the inside of assembly approximately 7/8 inch in from each sleeve and leg opening. Stitch, 1/8 to 3/16 inch from edge overlapping ends of the strip approximately 1 to 1 1/2 inches from the leg outseam and sleeve inseam. Turn leather strip to the aluminized side and stitch 1/8 to 3/16 inch along free edge and across the overlapped portion through all plies of material.	301	LSa-1 (each edge)	6-8	

Source: https://assist.dla.mil -- Downloaded: 2026-10-26T04:40Z
Check the source to verify that this is the current version before use.

TABLE I - CONSTRUCTION OF COVERALLS

No.	Operation	Stch Type	Seam/ Stch type	Stch/ in	Thread (3.3.4)
20.	Attach leather strips (cont'd)				
	NOTE: When splicing is used, the seam shall be spread open and flat (see operation 4e).				
	OR				
	As an option, fold leather strip in half lengthwise and using a folder, stitch each sleeve and leg opening with a single or double row of stitching 1/8 to 3/16 inch from edges of leather strip, thru all plies of material, overlapping ends of strip, approximately 1 to 1 1/2 inches from the sleeve inseam and leg outseam. The ends of the sleeves and legs shall extend within and to the folded edge of the leather strips.	301	BSa-1 or BSa-2	6-8	
21.	Make and attach hanger loop.				
	a. Fold hanger loop lengthwise in thirds and stitch along each folded edge 1/8 inch from the edge.	301 or 401	EFaa-2	6-8	
	b. Fold under ends of hanger loop and center loop on back facing about 1 inch below the neckline. Bartack or stitch and backstitch across the entire width of the ends through all plies.	Bar-tack or 301	LSd-1	22 to 28 per bar-tack or 12-16	
22.	Clean coverall.				
	a. Trim all ends of thread and remove all loose thread.				
	b. Remove all spots and stains.				
	NOTE: Care shall be taken in the use of the solvents for cleaning to assure that the coating on the aluminized material is not damaged.				

Source: https://assist.dla.mil -- Downloaded: 2026-10-26T04:40Z
Check the source to verify that this is the current version before use.

TABLE I - CONSTRUCTION OF COVERALLS

No.	Operation	Stch Type	Seam/ Stch type	Stch/ in	Thread (3.3.4)
19.	Make and attach inside protective flap and dee-rings. (cont'd)				
	b. Stitch 3/8 inch from the raw edges the top, bottom and the inside edges of the protective flap face to face with the liner. Trim corners, turn and topstitch 3/8 inch gauge. Overedge raw edges together.	301 or 401 503, 504 or 602	SSe-2; SSa-1	6-8	
	c. Position the aluminized side of the flap against the inside of the coverall at left front. Overlap the front about 1 inch with the top finishing 1/2 (± 1/8) inch below the neck edge, even with the top of the outside flap. Stitch flap to coverall about 1/4 inch from the overedged edge, parallel to the previously formed row of stitching on the coverall.	301	LSa-1		
	d. Rivet four dee-rings through the front edge of the outside protective flap to correspond with the snaps on the left front. The back inside edges of the dee-rings shall be at the edge of the flap, and when fastened, there shall be no twisting or bulging of the fronts.				
20.	Attach leather strips.				
	Insert a two-inch wide leather strip on the inside of assembly approximately 7/8 inch in from each sleeve and leg opening. Stitch, 1/8 to 3/16 inch from edge overlapping ends of the strip approximately 1 to 1 1/2 inches from the leg outseam and sleeve inseam. Turn leather strip to the aluminized side and stitch 1/8 to 3/16 inch along free edge and across the overlapped portion through all plies of material.	301	LSa-1 (each edge)	6-8	

Source: https://assist.dla.mil -- Downloaded: 2026-10-26T04:40Z
Check the source to verify that this is the current version before use.

TABLE I - CONSTRUCTION OF COVERALLS

No.	Operation	Stch Type	Seam/ Stch type	Stch/ in	Thread (3.3.4)
20.	Attach leather strips (cont'd)				
	NOTE: When splicing is used, the seam shall be spread open and flat (see operation 4e).				
	OR				
	As an option, fold leather strip in half lengthwise and using a folder, stitch each sleeve and leg opening with a single or double row of stitching 1/8 to 3/16 inch from edges of leather strip, thru all plies of material, overlapping ends of strip, approximately 1 to 1 1/2 inches from the sleeve inseam and leg outseam. The ends of the sleeves and legs shall extend within and to the folded edge of the leather strips.	301	BSa-1 or BSa-2	6-8	
21.	Make and attach hanger loop.				
	a. Fold hanger loop lengthwise in thirds and stitch along each folded edge 1/8 inch from the edge.	301 or 401	EFaa-2	6-8	
	b. Fold under ends of hanger loop and center loop on back facing about 1 inch below the neckline. Bartack or stitch and backstitch across the entire width of the ends through all plies.	Bar-tack or 301	LSd-1	22 to 28 per bar-tack or 12-16	
22.	Clean coverall.				
	a. Trim all ends of thread and remove all loose thread.				
	b. Remove all spots and stains.				
	NOTE: Care shall be taken in the use of the solvents for cleaning to assure that the coating on the aluminized material is not damaged.				

Source: https://assist.dla.mil -- Downloaded: 2026-10-26T04:40Z
Check the source to verify that this is the current version before use.

3.8 Sizes and measurements. The sizes and measurements of the finished coveralls shall be in accordance with Table II (see figure 1). All measurements and tolerances are expressed in inches.

Table II - Sizes and Finished Measurements

Size	(A) Chest	(B) Sleeve Inseam	(C) Leg Inseam	(D) Center Back Length
Small/Medium (37-41)	47	24	25 3/4	59 1/2
Large/X-Large (42-46)	52	25	26 3/4	61 1/2
Tolerances	± 1	± 1/2	± 1/2	± 1

NOTE: Measurements shall be taken with coveralls laid smooth and flat with fronts facing up and the outside protective flap completely closed. Refer to figure 1 for A, B, C, and D measurements.

(A) Chest - Twice the measurement taken across front at base of armholes from folded edge to folded edge.
(B) Sleeve Inseam - Measure inseam from bottom of armhole seam to bottom edge of leather bound sleeve opening.
(C) Leg Inseam - Measure inseam from crotch (junction of center front and center back seam) to bottom edge of leather bound leg opening.
(D) Center Back Length - Measure center back length from neckline to bottom edge of leather bound leg opening.

3.9 Workmanship. The finished coveralls shall conform to the quality of product established by this specification. The occurrence of defects shall not exceed the applicable point value or defect limit.

4. QUALITY ASSURANCE PROVISIONS

4.1 Responsibility for inspection. Unless otherwise specified in the contract, the contractor is responsible for the performance of all inspection requirements as specified herein. Except as otherwise specified in the contract or order, the contractor may use his own or any other facilities suitable for performance of the inspection requirements specified herein, unless disapproved by the Government. The Government reserves the right to perform any of the inspections set forth in the specification where such inspections are deemed necessary to assure supplies and services conform to prescribed requirements.

4.1.1 Certificate of compliance. Where certificates of compliance are submitted, the Government reserves the right to check test such items to determine the validity of the certification.

Source: https://assist.dla.mil -- Downloaded: 2026-10-26T04:40Z
Check the source to verify that this is the current version before use.

1/ Unless otherwise specified, a certificate of compliance may be accepted for this characteristic.
2/ When testing is specified, the silicone treated leather shall be immersed in water for 1/2 hour and tested in accordance with Test Method 8111 of FED-STD-311.

4.4.2 End item examination. Sampling and examination of the end item shall be performed in accordance with the provisions of MIL-STD-1666.

4.4.3 Packaging inspection. Packaging inspection shall be in accordance with MIL-STD-1666.

5. PACKAGING

5.1 Preservation-packaging. Preservation-packaging shall be level A or C as specified (see 6.2).

5.1.1 Level A. Each coverall shall be folded as follows:

1. Close slide fastener and fasten snaps.
2. Place coverall flat, face down.
3. Bring legs up even with the back neck line.
4. Fold sleeves at right angles across the back.
5. Fold coveralls in half across the chest. The completely folded coveralls shall measure approximately 23 by 14 1/2 inches. Each coverall shall be inserted in a snug-fitting flat style clear polyethylene film bag of not less than 0.003 inch thickness. All seams and closures shall be formed by heat sealing. A 1/4 inch diameter hole shall be made at one corner of each polyethylene bag to allow excess air to escape. As an alternate, the polyethylene bag may be of the tuck-in or reverse flap type, in which a heat seal closure and corner vent hole are not required.

5.1.2 Level C (Commercial packaging). Coveralls shall be packaged to afford adequate protection against deterioration and physical damage during shipment from the supply source to the first receiving activity. The package and the quantity per package shall be the same as that normally used by the contractor for retail distribution.

5.2 Packing. Packing shall be level A, B, or C as specified (see 6.2).

* 5.2.1 Level A. Five coveralls of one size only, packaged as specified in 5.1, shall be packed in a fiberboard shipping container assembled, closed and reinforced conforming to type CF, class weather resistant, grade V15C, variety DW, size 3A of MIL-B-17757. The fiberboard for the liner shall conform to type CF, class domestic of MIL-B-17757. Level A packages shall be packed flat within the shipping container with every other coverall reversed end to end and the length of the coverall parallel to the side panels of the container. Each container shall have the contents completely covered on the top and bottom with a sheet of commecial grade kraft paper. Toward the end of the contract, or when there are less than the required amount per container of the same size, mixed sizes may be packed within the same shipping container.

Source: https://assist.dla.mil -- Downloaded: 2026-10-26T04:40Z
Check the source to verify that this is the current version before use.

1/ Unless otherwise specified, a certificate of compliance may be accepted for this characteristic.
2/ When testing is specified, the silicone treated leather shall be immersed in water for 1/2 hour and tested in accordance with Test Method 8111 of FED-STD-311.

4.4.2 End item examination. Sampling and examination of the end item shall be performed in accordance with the provisions of MIL-STD-1666.

4.4.3 Packaging inspection. Packaging inspection shall be in accordance with MIL-STD-1666.

5. PACKAGING

5.1 Preservation-packaging. Preservation-packaging shall be level A or C as specified (see 6.2).

5.1.1 Level A. Each coverall shall be folded as follows:

1. Close slide fastener and fasten snaps.
2. Place coverall flat, face down.
3. Bring legs up even with the back neck line.
4. Fold sleeves at right angles across the back.
5. Fold coveralls in half across the chest. The completely folded coveralls shall measure approximately 23 by 14 1/2 inches. Each coverall shall be inserted in a snug-fitting flat style clear polyethylene film bag of not less than 0.003 inch thickness. All seams and closures shall be formed by heat sealing. A 1/4 inch diameter hole shall be made at one corner of each polyethylene bag to allow excess air to escape. As an alternate, the polyethylene bag may be of the tuck-in or reverse flap type, in which a heat seal closure and corner vent hole are not required.

5.1.2 Level C (Commercial packaging). Coveralls shall be packaged to afford adequate protection against deterioration and physical damage during shipment from the supply source to the first receiving activity. The package and the quantity per package shall be the same as that normally used by the contractor for retail distribution.

5.2 Packing. Packing shall be level A, B, or C as specified (see 6.2).

* 5.2.1 Level A. Five coveralls of one size only, packaged as specified in 5.1, shall be packed in a fiberboard shipping container assembled, closed and reinforced conforming to type CF, class weather resistant, grade V15C, variety DW, size 3A of MIL-B-17757. The fiberboard for the liner shall conform to type CF, class domestic of MIL-B-17757. Level A packages shall be packed flat within the shipping container with every other coverall reversed end to end and the length of the coverall parallel to the side panels of the container. Each container shall have the contents completely covered on the top and bottom with a sheet of commecial grade kraft paper. Toward the end of the contract, or when there are less than the required amount per container of the same size, mixed sizes may be packed within the same shipping container.

Source: https://assist.dla.mil -- Downloaded: 2026-10-26T04:40Z
Check the source to verify that this is the current version before use.

6.2 Ordering data. Procurement documents should specify the following:

a. Title, number and date of this specification.
b. Sizes required (see 1.2).
c. When first article inspection is required (see 3.2). (When required, the item tested should be a first article sample. The contracting officer should include specific instructions in acquisition documents regarding arrangements for examination, quantity, and testing and approval of the first article.)
d. Selection of applicable levels of packing and packaging (see 5.1 and 5.2).
e. List of approved silicone compounds (see 3.3.3.1).

6.3 Samples and patterns. For access to samples and patterns, address the procuring activity issuing the invitation for bids.

6.4 Silicone compounds and flame resistant treatments. Only those silicone compounds and flame resistant treatments approved by the Natick Army Research and Development Center, Natick, MA 01760, and the appropriate medical service and so listed in the invitation for bids or request for proposal, shall be considered acceptable for the related procurement (see 3.3.3 and 3.3.3.1).

* 6.5 Changes from previous issue. The margins of this specification are marked with an asterisk to indicate where changes (additions, modifications, corrections, deletions) from the previous issue were made. This was done as a convenience only and the Government assumes no liability whatsoever for any inaccuracies in these notations. Bidders and contractors are cautioned to evaluate requirements of this document based on the entire content irrespective of the marginal notations and relationship to the last previous issue.

Custodians:
Navy - NU

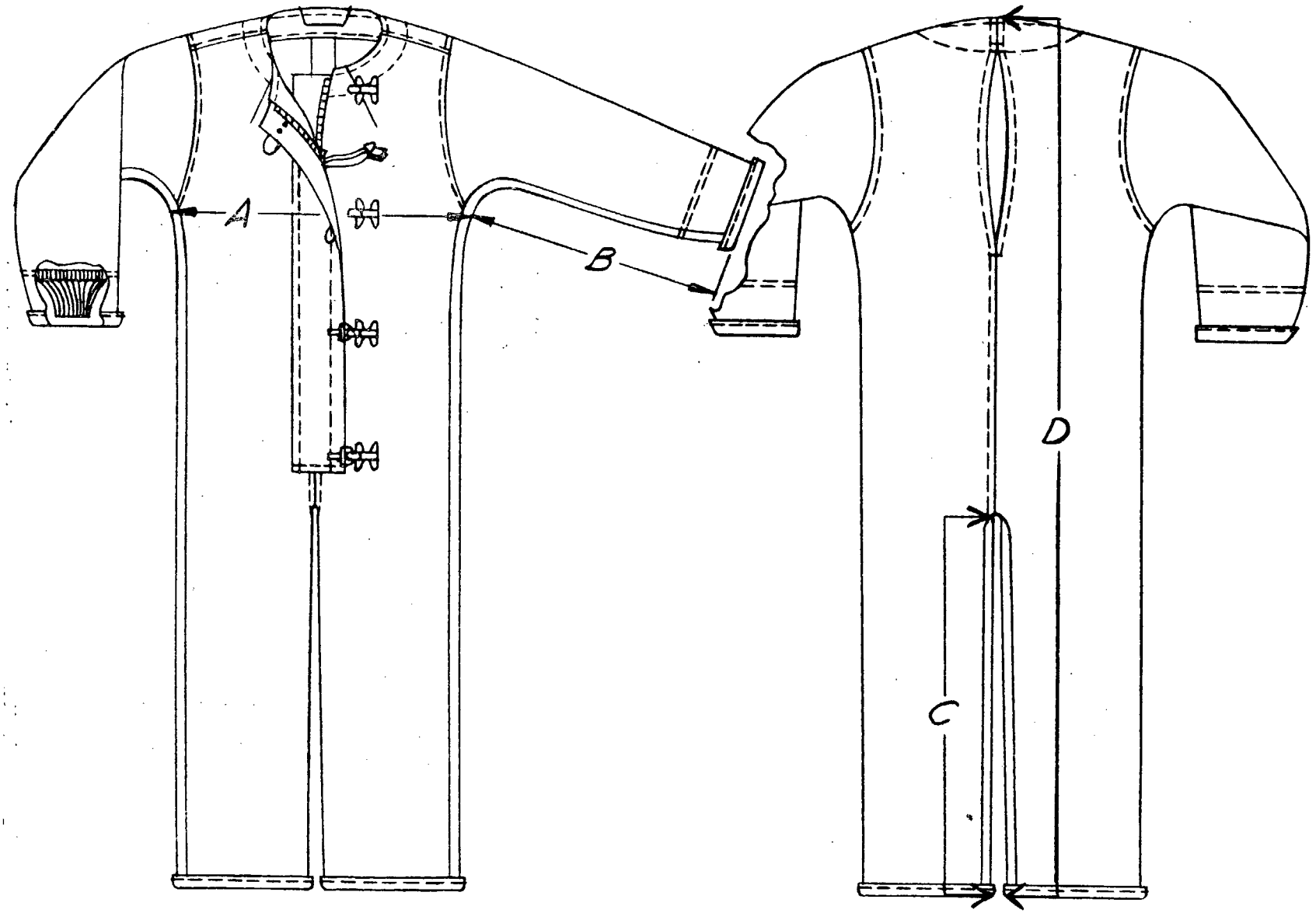
Preparing Activity:
Navy - NU

Review Activity:
Navy - SH
DLA - CT

Project No. 8415-0385

User Activity:
Navy - AS, CG, MC

Source: https://assist.dla.mil -- Downloaded: 2026-10-26T04:40Z
Check the source to verify that this is the current version before use.

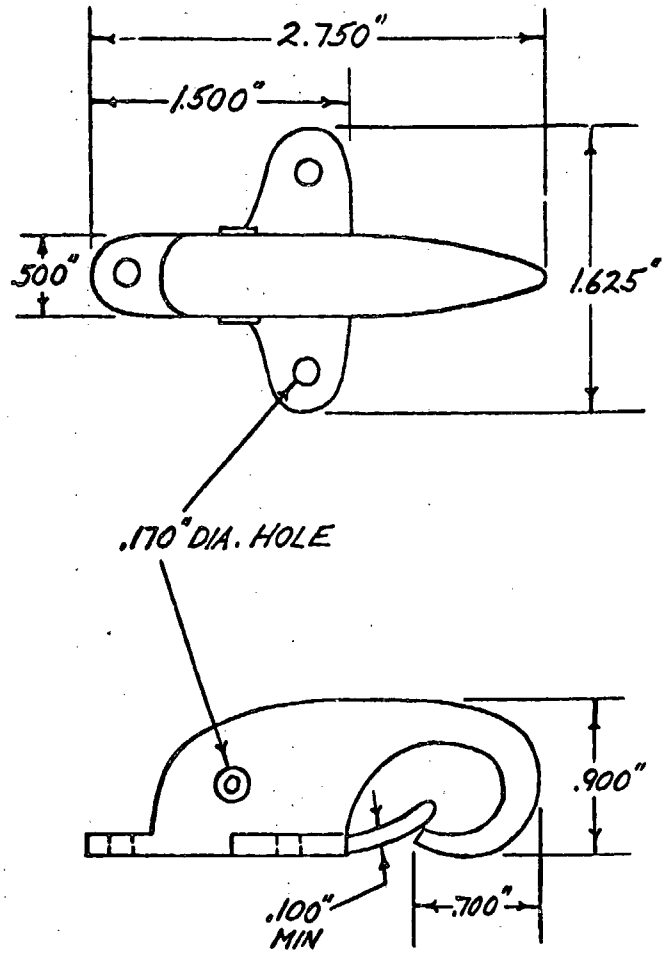


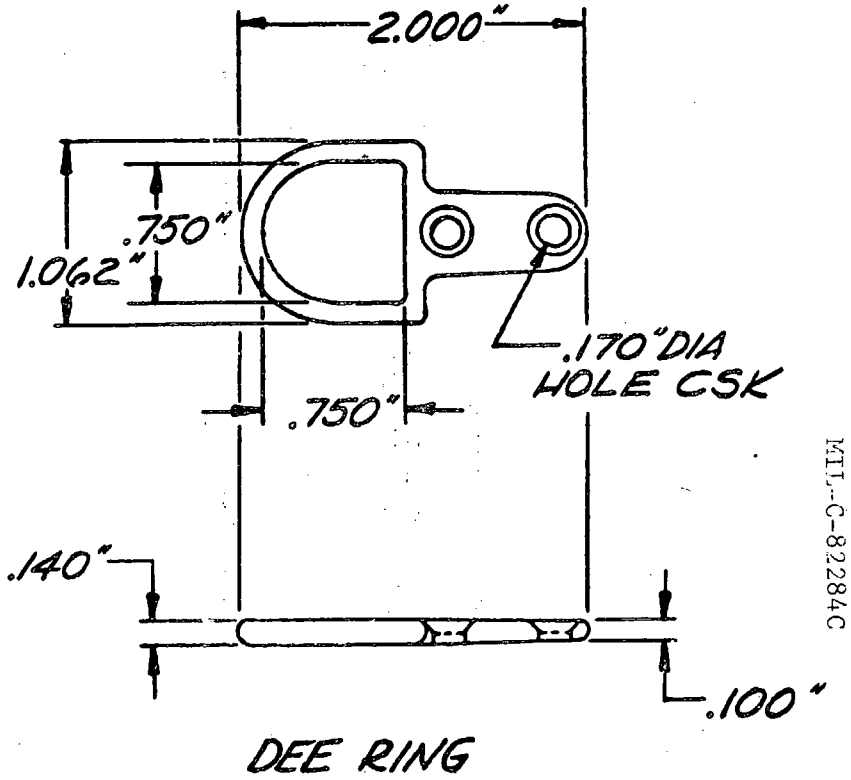
FRONT

BACK

FIG. 1 COVERALLS, FIREMAN'S, ALUMINIZED, PROXIMITY

Source: https://assist.dla.mil -- Downloaded: 2026-10-26T04:40Z
Check the source to verify that this is the current version before use.





TOLERANCE ± .010 INCH

FIGURE 2 SNAP AND DEE RING

Source: https://assist.dla.mil -- Downloaded: 2026-10-26T04:40Z
Check the source to verify that this is the current version before use.

INSTRUCTIONS: In a continuing effort to make our standardization documents better, the DoD provides this form for use in submitting comments and suggestions for improvements. All users of military standardization documents are invited to provide suggestions. This form may be detached, folded along the lines indicated, taped along the loose edge *(DO NOT STAPLE)*, and mailed. In block 5, be as specific as possible about particular problem areas such as wording which required interpretation, was too rigid, restrictive, loose, ambiguous, or was incompatible, and give proposed wording changes which would alleviate the problems. Enter in block 6 any remarks not related to a specific paragraph of the document. If block 7 is filled out, an acknowledgement will be mailed to you within 30 days to let you know that your comments were received and are being considered.

NOTE: This form may not be used to request copies of documents, nor to request waivers, deviations, or clarification of specification requirements on current contracts. Comments submitted on this form do not constitute or imply authorization to waive any portion of the referenced document*(s)* or to amend contractual requirements.

(Fold along this line)

(Fold along this line)

DEPARTMENT OF THE ARMY

NO POSTAGE NECESSARY IF MAILED IN THE UNITED STATES

OFFICIAL BUSINESS
PENALTY FOR PRIVATE USE $300

BUSINESS REPLY MAIL
FIRST CLASS PERMIT NO. 12062 WASHINGTON D. C.
POSTAGE WILL BE PAID BY THE DEPARTMENT OF THE ARMY

Commander
US Army Natick Research and Development
Laboratories
ATTN: DRDNA-ES (Mr. D. Luppino)
Natick, MA 01760

Source: https://assist.dla.mil -- Downloaded: 2026-10-26T04:40Z
Check the source to verify that this is the current version before use.

STANDARDIZATION DOCUMENT IMPROVEMENT PROPOSAL

(See Instructions – Reverse Side)

1. DOCUMENT NUMBER	2. DOCUMENT TITLE

3a. NAME OF SUBMITTING ORGANIZATION	4. TYPE OF ORGANIZATION *(Mark one)*
	☐ VENDOR ☐ USER ☐ MANUFACTURER ☐ OTHER *(Specify)*: _____
b. ADDRESS *(Street, City, State, ZIP Code)*	

5. PROBLEM AREAS

a. Paragraph Number and Wording:

b. Recommended Wording:

c. Reason/Rationale for Recommendation:

6. REMARKS

7a. NAME OF SUBMITTER *(Last, First, MI)* – Optional	b. WORK TELEPHONE NUMBER *(Include Area Code)* – Optional
c. MAILING ADDRESS *(Street, City, State, ZIP Code)* – Optional	8. DATE OF SUBMISSION *(YYMMDD)*

DD FORM 82 MAR 1426

PREVIOUS EDITION IS OBSOLETE.

Source: https://assist.dla.mil -- Downloaded: 2026-10-26T04:40Z
Check the source to verify that this is the current version before use.